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College of Information Technology Education

CAPSTONE PROJECT PROPOSAL

Custom Cake Ordering Platform with Smart Scheduling and Sales Analytics

In partial fulfillment for the requirements in Capstone 1 and Research Subject

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CHAPTER I

INTRODUCTION

I. Background of the Study

In today's fast-paced world, businesses across all sectors have shifted to digital platforms to keep up with the growing expectations of modern consumers [1]. The food service and baking industry in the Philippines experienced this exact transition, as tech-savvy individuals increasingly rely on mobile devices and computers to browse menus, place orders, and settle transactions remotely [2]. Because of this shift in consumer behavior, small local micro-enterprises that continue to rely on manual, paper-based processes face severe operational challenges. Traditional, non-automated methods often led to critical inaccuracies, limited market reach, and an inability to meet daily customer expectations. For small community bakeries, operating without a modern web presence means losing potential sales to larger competitors who offer instant online ordering, clear pricing, and transparent communication [1].

The core issue stems from how order information is captured, stored, and processed during daily operations [3]. In a digital commerce platform, data entries are structured, validated, and processed automatically by backend logic. Conversely, traditional store setups rely heavily on loose paper slips, handwritten notebooks, and human memory. This lack of automated validation leaves daily transactions highly vulnerable to misplaced records, delayed order tracking, and operational confusion. When a small business operates entirely without digital validation, everyday tasks like taking an order, calculating a price, or scheduling a delivery turn into stressful, repetitive manual chores that consume huge amounts of time and lower staff productivity. For community-scale enterprises, continuing to operate within a purely analog framework creates a technical barrier that cuts them off from the modern digital marketplace, making it increasingly difficult to sustain consistent growth or build long-term customer loyalty [3].

C&N Bakery is a local establishment serving a growing customer base within a competitive community market. The bakery offers a wide selection of daily baked goods, including fresh breads, pastries, cakes, and traditional local delicacies. In addition to selling standard retail items over the counter, the bakery specializes in custom cake orders made for special events like birthdays, weddings, debuts, and anniversaries. These custom orders represent a significant portion of the business's overall revenue and require careful planning and precise execution. Despite managing a diverse product menu and handling many daily customers, C&N Bakery handled its entire workflow through manual mechanisms, such as walk-in counter visits, phone calls, and disorganized social media messages on Facebook. The business operated without a centralized digital system to organize its daily activities, leaving crucial customer choices and transaction records disorganized across different notebooks, paper slips, and informal message threads.

When a business operates without a single central system to store its records, information easily gets lost or fragmented [4]. At C&N Bakery, a single custom transaction often left an unstructured data trail across multiple communication channels. Design layouts, delivery deadlines, special instructions, and payment receipts were frequently recorded in completely different places. Because these details were isolated from one another, staff members had to manually piece together the customer's requirements before baking could start. This manual approach consumed massive amounts of time and made day-to-day operations chaotic, turning simple routine tasks

into exhausting manual procedures that drained the small business's daily energy and workforce productivity.

Managing custom cake orders is inherently far more complex than selling pre-baked goods over the counter [4]. A single custom cake request requires capturing and tracking multiple precise attributes simultaneously, including cake dimensions, tier structures, flavor variations, frosting types, color schemes, personalized dedication messages, and visual reference photos. At the same time, the bakery must carefully coordinate pickup or delivery dates to guarantee product freshness and structural integrity during transit. Because C&N Bakery relied on handwritten logs, loose paper receipts, and unorganized chat histories, staff members easily misread or forgot specific design preferences by the time production began in the kitchen. Furthermore, writing down fulfillment dates manually in physical logbooks frequently caused human error, resulting in accidental double-bookings that overloaded the kitchen team beyond their daily shift capacity.

This operational vulnerability is deeply tied to the problem of manual transcription [3]. When a customer outlines their custom selections through casual communication channels, that information exists as an unformatted text string. As that request moves from the front counter staff to the production area, the information is manually rewritten onto paper order tags or kitchen whiteboards. Every time a human manually copies, interprets, or rewrites an order detail, the risk of human error occurring increases significantly. Essential design choices can easily be misread, miscalculated, or completely forgotten during the transfer. This manual workflow created a weak operational loop where the original intent of the customer rarely matched the actual physical instructions given to the baking staff on the kitchen floor.

Every time an order detail is manually transcribed, the risk of miscalculation increases. Essential design preferences can easily be misinterpreted, miscalculated, or left out entirely. When a cake is prepared with incorrect specifications, it triggers a chain reaction of operational failures across the business, resulting in wasted raw ingredients, disrupted baking schedules, lost revenue, and damaged customer trust. The financial impact of these errors is particularly severe due to the rigid nature of the baking process itself. Unlike quick-service meals that can be replaced in minutes, custom cakes require hours of uninterrupted baking, cooling, and detailed decorating. If an error is noticed late in the process, the entire cake often has to be discarded, delaying every other order waiting in the kitchen queue and creating a backlog that ruins the entire work shift.

These internal difficulties were further magnified by the rapid adoption of digital payment solutions in the Philippines, particularly electronic wallets like GCash and online bank transfers [5]. Modern consumers expect clear pricing, immediate feedback, and simple remote payment options. At C&N Bakery, operating manually created major communication bottlenecks. When a customer sent a message asking for a price quote on a custom cake design, kitchen staff had to pause their physical work to manually calculate costs line-by-line based on cake size, tiers, and add-ons. This manual step led to long response delays, leaving potential buyers waiting in digital queues for simple information. In an era where buyers expect instant answers, waiting hours for a message reply caused many customers to abandon their inquiries and purchase from larger, automated competitors instead [5].

Furthermore, once an order was placed, customers lacked a clear way to monitor their order status in real time. Because there was no online order tracker, customers were forced to call or message the store repeatedly to ask if their cake was being prepared, baked, or ready for pickup. This constant back-and-forth communication gap created a major operational barrier inside the store. Front counter staff and kitchen personnel did not share a synchronized information system, making it hard to track order priorities and preparation sequences. Staff members constantly stopped their physical duties to verbally verify order statuses with the bakers. This workplace interruption wasted valuable labor time, stressed out the staff, and led to scheduling mistakes that left waiting customers frustrated.

Operating without a secure web system also introduced serious data security risks and administrative obstacles [3].

On the customer side, users lacked a secure account interface to review their past orders or monitor active deliveries. For administrative personnel, operating without a secure, centralized administrative dashboard meant keeping financial records, customer contact details, and transaction histories in paper notebooks stored directly inside the active kitchen. These paper logs were constantly exposed to water spills, flour dust, grease stains, heat degradation, and physical loss. If a single notebook was misplaced, damaged, or destroyed, the bakery permanently lost its operational records with no way to recover the data. Storing customer addresses and phone numbers on loose paper slips also presented privacy concerns, while management was left without historical records to evaluate monthly revenue, track sales growth, or identify best-selling product trends.

To resolve these operational vulnerabilities, this study will develop the Web-Based Order, Tracking, Customization, and Sales Analytics System for C&N Bakery. The proposed web application will establish a centralized digital platform engineered to streamline order taking, dynamically balance daily kitchen capacity limits, track production stages across a clear lifecycle, and compile historical records into clear visual sales reports. The system will provide dedicated user accounts for registered customers while maintaining open browsing access for guest users.

Through this proposed platform, customers will be able to customize their cake selections, view automated cost calculations, submit reference images, and select available fulfillment dates backed by real-time schedule checks to prevent kitchen overbooking. Administrative staff will be provided with a centralized management dashboard to verify order payments, update production progress in real time, and view visual sales analytics. Built using cost-effective cloud database technology, this practical architecture will keep operational expenses manageable, safeguard store transaction records, eliminate manual data entry errors, optimize kitchen productivity, and provide C&N Bakery with an efficient, reliable web presence tailored to its daily business needs.

1.2 Statement of the Problem

The main problem facing C&N Bakery was its reliance on paper notebooks, wall calendars, and chat messages on social media to handle daily operations [3]. Because the business did not have a single web application, store records were scattered across loose paper slips and messaging apps, causing several day-to-day workflow problems:

1. **Slow Replies and Unclear Pricing:** Customers often waited hours to get a response about prices or available cakes because store staff had to answer online messages while attending to walk-in customers [1, 5]. In addition, custom cake prices were calculated by hand without standard formulas, which led to incorrect price quotes [3].
2. **Order Copying Errors and Wasted Materials:** Staff had to manually rewrite custom cake details from chat apps onto paper kitchen slips [4]. This manual step frequently caused staff to misread cake colors, tier sizes, or dedication messages, resulting in wrong cake designs, unhappy customers, and wasted ingredients [3, 4].
3. **Kitchen Overbooking and Disorganized Workflows:** Order pickup dates were tracked on a paper wall calendar without any automatic system to limit daily orders [4]. As a result, the kitchen team often accepted too many orders, leading to rushed work and late pickups [4]. Staff also had to constantly stop bakers while they worked just to ask for order status updates [3].
4. **Pickup Scheduling Mix-ups and Missing Records:** Using handwritten pickup notes led to misread dates, causing cakes to be baked too early or forgotten completely until the customer arrived [4]. Paper sales slips were also easily lost or misplaced, leaving gaps in the bakery's daily record books [3].
5. **Unsecured Records and Lack of Sales Reports:** Customer phone numbers, home addresses, and sales notes were kept in paper logs kept in the kitchen, where they could easily be damaged by water or flour spills [3]. The bakery owner also had to manually add up paper receipts to check monthly earnings, leaving

no quick way to track profits or best-selling products [1, 3].

Without a centralized web application that includes custom cake order forms, automated kitchen scheduling, order status tracking, and sales reporting tools, C&N Bakery will continue to face daily operational errors, misplaced records, and lost sales opportunities.

1.3 Objectives of the Study

The primary objective of this project is to design and develop an easy-to-use web platform for C&N Bakery to simplify ordering, organize kitchen schedules, and track store performance.
Specifically, this study aims to:

1. Understand Store Needs: Study the current paper-based processes, order delays, and kitchen limits at C&N Bakery to determine the core requirements for the web platform.
2. Design User-Friendly Workflows: Create clear layout designs, smooth navigation flows, and database structures for customer ordering, schedule checks, order tracking, and sales monitoring.
3. Develop the Web Platform: Build a responsive, easy-to-navigate website that helps customers place custom orders, checks pickup availability, and displays live order status updates.
4. Create an Admin Reporting Dashboard: Build an easy-to-read management dashboard that organizes store transactions into visual sales summaries and store records.
5. Test and Improve the System: Test the web app for speed, ease of use, and overall accuracy by gathering direct feedback from bakery staff and test customers.

1.4 Scope and Delimitation of the Study

This study focuses on designing, developing, and evaluating a web-based ordering, tracking, and management platform for C&N Bakery to transition its daily store operations into a centralized digital system. The project aims to minimize reliance on handwritten logs, paper calendars, and unorganized social media message threads by deploying a responsive web application that serves both store customers and administrative personnel.

A. System Setup and Accessibility

The platform will be developed as a responsive, mobile-friendly web application, eliminating the need for client-side app installations. The system interface will dynamically adapt across desktop computers, laptops, tablets, and smartphones, operating on standard modern web browsers (e.g., Google Chrome, Mozilla Firefox, Apple Safari). The application will require an active internet connection to communicate with a centralized cloud database in real time.

B. User Roles and Access Controls

The study defines two primary user roles for system interactions:

- **Bakery Customers:** Unregistered guest users will be able to browse available menu items, product descriptions, FAQs, and customer reviews. Registered customers will have access to a personal portal to customize cake requests, submit orders, attach reference images, and track active order statuses.
- **Administrative and Kitchen Personnel:** Store management and kitchen staff will share a secure backend administrative module to oversee daily operations. This includes reviewing incoming order details, monitoring kitchen schedules, updating baking progress, adjusting menu item availability, setting daily order capacity caps, and viewing automated sales summaries.

C. Scope of System Capabilities

The core functionality of the web platform will cover the following operational areas:

- **Customer Interface:** Provides categorized product browsing, interactive custom cake design selections, reference photo attachment, automated schedule checks to block fully booked pickup dates, shopping cart management, live order progress tracking, automated email updates, past order history, user ratings, and basic account profile management.
- **Administrative & Kitchen Management:** Provides a central dashboard displaying daily store summaries, order search and filtering tools, detailed order views to verify design specifications and payment attachments, stage-by-stage baking status updates, product catalog and price management, store availability toggles, kitchen calendar capacity controls, and exportable visual sales reporting.

D. Delimitations of the Study

To keep the project scope manageable and tailored to C&N Bakery's operational requirements, the following boundaries are explicitly established:

1. **No Native Mobile Applications:** The system will be strictly built and deployed as a responsive web application and will not involve native mobile apps for app store distribution.
2. **No Automated Payment Gateway APIs:** The system will not integrate with direct banking APIs or third-party automated payment processors. Payment verification will rely on manual administrative review of customer-uploaded payment screenshots (e.g., GCash or bank transfers).
3. **No Third-Party Courier Integration:** The platform will not connect to external delivery services or logistics APIs. Coordinating delivery personnel or driver dispatch remains a manual task handled outside the web application.
4. **No Raw Ingredient Inventory Management:** The software will not track raw ingredient stock weights (such as flour, sugar, or eggs). Stock control within the system will rely on manual item toggles by store staff.
5. **No HR, Payroll, or Supplier Management:** The platform focuses exclusively on order handling, kitchen scheduling, and sales analytics; it will not track employee hours, calculate payroll, or manage ingredient suppliers.
6. **No Offline Mode:** The system relies on a cloud database infrastructure and requires active internet access for both customer and administrative functions.
7. **No Automated Refunds or Post-Confirmation Cancellations:** The platform will not execute direct monetary refunds or process automated cancellations once an order has been officially confirmed by store staff.

1.5 Significance of the Study

Developing the web-based ordering, tracking, and management platform for C&N Bakery will bring practical value to everyone involved in the daily operations and growth of the business. As local food enterprises adapt to modern commercial trends, community bakeshops require reliable digital tools to maintain efficiency and competitiveness. This study illustrates how a neighborhood bakery can transition away from traditional paper-based workflows. By replacing manual notebooks, physical calendars, and unorganized social media message threads with a centralized web platform, this project establishes a structured order handling process. It provides transparent pricing for customers, protects kitchen staff from daily overbooking, and safeguards store records from physical damage or loss.

this study provides clear, long-term benefits to the following stakeholders:

1. Bakery Customers and Guests

Traditionally, buyers placing custom orders face delayed communication, missing order details, and unclear price estimates. The proposed platform directly benefits customers by providing upfront price transparency through clear

cost breakdowns prior to order confirmation, effectively eliminating price confusion. Additionally, buyers can specify custom order details directly within the system to ensure final products align with their exact specifications, monitor order progress conveniently through the web platform without waiting for manual chat responses, and browse menu offerings from any internet-connected device.

2. Production and Kitchen Staff

Kitchen personnel managing manual logs face daily pressure coordinating fulfillment schedules and specific order requirements. This project organizes kitchen workflows by applying daily capacity limits to prevent kitchen staff from being overloaded beyond manageable shift capacity. Staff can access consolidated order specifications on a single screen, reducing mistakes caused by searching through loose paper notes or chat threads. Furthermore, personnel can update order progress step-by-step to maintain a clear workflow from preparation to final pickup, while quickly modifying item availability on the website whenever ingredients run short.

3. Executive Management (The Bakery Owner)

Managing a custom bakery without digital records complicates store administration and financial tracking. The platform assists management by consolidating customer transactions and payment receipts into structured digital records, minimizing manual calculation errors. Store management can review aggregated sales summaries to observe revenue trends and top-performing products, which simplifies inventory planning and budgeting. Moreover, built-in access controls protect store transaction histories and customer contact information from unauthorized access, assisting management in making informed decisions regarding product catalog adjustments, pricing models, and promotional strategies.

4. C&N Bakery as an Enterprise

Transitioning to a centralized web application enhances operational reliability and protects business reputation. The store benefits from reduced material waste, as clear order records and verified payment confirmations help avoid wasting raw baking ingredients on unconfirmed or misread orders. Providing structured digital ordering also gives the bakery a competitive advantage over non-automated local competitors and allows the business to manage higher sales volumes during holiday seasons without chaotic workflows or missed fulfillment deadlines.

5. Future Software Developers

Standard retail web applications often do not accommodate custom order workflows or daily kitchen scheduling constraints. This study provides future developers with a practical architectural reference for structuring dynamic web forms that handle custom pricing based on user-selected parameters. It also serves as a technical baseline for implementing schedule-balancing logic that restricts order bookings based on daily operational capacity.

6. Future Academic Researchers

This study provides clear documentation on how a small retail business shifts from manual operations to a web-based system. It serves as a practical reference for researchers studying business process automation and offers a baseline for future student research, such as expanding system capabilities to include automated SMS notifications or third-party logistics integrations.

7. The Local Food Service Industry

This project demonstrates that practical, cost-effective web applications are accessible to small food vendors without requiring extensive software budgets. Demonstrating how digital tools improve operational efficiency encourages other neighborhood micro-enterprises to modernize their business models and adapt to evolving local market demands.

8. BSIT Students

For Information Technology students, this study provides a practical guide for analyzing manual business problems and translating them into functional software modules. It illustrates how real-world store operations are converted into structured database models, intuitive user interfaces, and role-based security controls.

1.6 Definition of Terms

- Web-Based System: A central online software application running on a cloud server that allows customers, store staff, and management to handle ordering, tracking, and sales management through standard web browsers without installing local applications [3].
- User Authentication and Access Control: A security process that checks user login credentials and enforces account permission limits, automatically directing customers to the storefront, staff to order processing, and management to sales reports [3, 4].
- Interactive Custom Cake Builder: An online customization interface where buyers choose specific cake parameters, such as tier counts, sizes, flavors, and frostings, while adding custom text notes and reference photos [4].
- Automated Price Estimation Module: A background calculation script that instantly updates total cost estimates as customers select different cake options, eliminating manual price calculation delays over chat messages [3].
- Online Product Catalog: An organized digital display that categorizes C&N Bakery's product menu—such as cakes, cupcakes, and local delicacies—into clear, searchable groups for easy browsing [1, 2].
- Customer Portal and Status Tracker: A personal account dashboard where registered customers can review past purchase histories and monitor the current progress stage of active custom orders [2, 4].
- Sales Analytics Reporting: A management reporting tool that compiles stored sales database records into visual charts to help the bakery owner track monthly revenue growth and top-selling items [1, 3].
- Product Availability Control: An administrative management setting that lets bakery staff quickly hide specific cake choices or menu items whenever ingredients are out of stock [3].
- Capacity-Based Schedule Control: An automated calendar rule that tracks daily custom cake orders and blocks full pickup dates once the bakery reaches its maximum daily kitchen capacity [4].

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CHAPTER II

REVIEW OF RELATED LITERATURE

2.1 Theoretical Foundations of Web-Based Bakery Ordering and Real-Time Sync

Modern web-based e-commerce frameworks rely on event-driven architectures and real-time database synchronization to streamline retail operations [1], [3]. Traditional manual order intake dependent on physical logbooks or unstructured social media messages introduces significant latency, pricing errors, and communication gaps between front-desk staff and kitchen operations [2], [3].

In custom order environments, total order pricing P_{total} is dynamically calculated by aggregating base tier costs, volumetric flavor multipliers, and decorative component add-ons [2]:

$$P_{total} = P_{base} + \sum_{i=1}^{m} C_{flavor,i} + \sum_{j=1}^{k} C_{decor,j} + C_{fulfillment}$$

Where P_{base} represents the baseline cost determined by tier dimensions, C_{flavor} denotes specialty cake/frosting costs across m tiers, C_{decor} accounts for k custom decorative elements, and $C_{fulfillment}$ incorporates delivery or specialized handling fees [2]. Executing this pricing calculation in-browser eliminates manual arithmetic mistakes and immediately calculates the required 50% checkout deposit [2].

Figure 2.1: Event-driven real-time database synchronization architecture for web-based bakery operations [1], [4].

Furthermore, real-time database websockets enable instant status updates across production workflows [1], [2]. As kitchen staff transition orders through discrete stages (*Preparing* → *Baking* → *Packing* → *Ready*), updates synchronize directly to the client interface within $< 2seconds$, eliminating the need for periodic manual customer inquiries [1], [2].

2.2 Comparison of Bakery Management and Scheduling Approaches

Management Paradigm	Operational Mechanism	Hardware / Infrastructure Overhead	Hardware / Infrastructure Overhead
Manual Paper System	Paper order slips, physical	Minimal hardware; high	High rate of transcription

[3]	logbooks, and phone calls	physical paper	errors; susceptible to kitchen double-booking [2], [4
Social Media Messaging [2]	Direct messaging (e.g., Messenger) with manual screenshot tracking	Low hardware; relies on personal mobile devices	Disorganized design references; high risk of missed order specifications [2]
Cloud Web & Smart Scheduling Engine [1], [4]	Web-based configurator with Firebase Realtime Database and dynamic calendaRlocking	Very low; hosted on serverless cloud CDN and edge networks	High customization precision; automated dynamic slot locking preventing overbooking [2], [4]

Management Paradigm	Operational Mechanism	Hardware / Infrastructure Overhead	Customization & Capacity Control
Manual Paper System [3]	Paper order slips, physical logbooks, and phone calls	Minimal hardware; high physical paper and storage footprint	High rate of transcription errors; susceptible to kitchen double-booking [2], [4]
Social Media Messaging [2]	Direct messaging (e.g., Messenger) with manual screenshot tracking	Low hardware; relies on personal mobile devices	Disorganized design references; high risk of missed order specifications [2]

Cloud Web & Smart Scheduling Engine [1], [4]	Web-based configurator with Firebase Realtime Database and dynamic calendar locking	Very low; hosted on serverless cloud CDN and edge networks	High customization precision; automated dynamic slot locking preventing overbooking [2], [4]

2.3 Research Gap

Existing commercial software solutions for small-scale bakeries either over-complicate store workflows with high enterprise software fees or rely on basic online storefronts that lack dynamic daily capacity management [2], [3], [4]. On the other hand, relying strictly on informal social media messaging often leads to unorganized cake design notes, miscalculated prices, lost receipts, and severe overbooking during peak holiday seasons [2], [3], [4].

TRADITIONAL GAP / VULNERABILITIES [2], [3]	**PROPOSED PLATFORM SOLUTION [1], [4]**
- Informal chat **- Manual arithmetic errors** **- Double-booked kitchen logs** **- Paper receipt data loss**	**- Structured Cake Builder** **- Automatic Pricing Engine** **- Smart Dynamic Capacity Lock** **- Real-Time Cloud Analytics [5]**

Figure 2.3: Operational research gap and targeted system resolution framework [1], [2], [3], [4], [5].

This capstone project addresses this gap by developing a localized, cloud-integrated web application tailored for C&N Bakery [1], [2]. By combining an interactive custom cake configurator, dynamic calendar slot-locking via real-time database thresholds (dayLimits), a 4-stage kitchen production tracker, and automated sales analytics into a single low-cost system, the platform solves core operational vulnerabilities without burdening the bakery with high software costs or complex setups [1], [2], [4], [5].

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CHAPTER III

METHODOLOGY

3.1 Research Design

In this study, the research team utilizes a descriptive-developmental research design. This practical, two-phase approach combines two distinct methodologies:

1. The Descriptive Phase: Observing, analyzing, and documenting how C&N Bakery manages its daily operations using manual mechanisms, such as handwritten logbooks, wall calendars, loose paper slips, and physical paper receipts.
2. The Developmental Phase: Planning, modeling, constructing, testing, and refining a web-based ordering and management platform to address those specific operational challenges.

By linking these two phases, the research team ensures that every database table, system input field, and administrative tool addresses documented operational problems identified during field observations rather than speculative requirements.

DESCRIPTIVE-DEVELOPMENTAL RESEARCH DESIGN

DESCRIPTIVE PHASE	DEVELOPMENTAL PHASE
● Observed walk-in & phone orders ● Documented paper logbook issues ● Identified wall calendar capacity risks ● Recorded manual math errors	System Infrastructure & Security ● Cloud database, Vercel deployment & email alerts Customer Portal & Ordering System ● Custom cake builder & daily order limits Admin & Kitchen Management System ● Sales analytics & catalog management controls

Fig. 3.1. Descriptive-Developmental Research Methodology Framework.

3.1.1 Descriptive Research

The descriptive phase focused on observing and documenting how C&N Bakery operated in its daily store environment. Rather than setting up artificial tests, the research team spent time inside the bakery to observe how orders were taken, processed, tracked, and recorded.

Before writing software code, the research team visited C&N Bakery to analyze its manual workflow. The team interviewed the business owner, spoke with counter personnel, observed kitchen operations, and examined physical order records. These field observations revealed four main operational challenges:

1. Disorganized Customer Messages and Order Notes: Customer inquiries and custom cake requests arrived across multiple unlinked channels, including personal phone calls, SMS text messages, and chat threads on the bakery's Facebook page. Without a central record, counter staff had to scroll through long message histories on personal smartphones during busy shop hours to locate reference photos, price quotes, or pickup agreements.
2. Risks of Handwritten Paper Order Slips: Custom cake details—such as tier dimensions, flavor choices, frosting types, color themes, and dedications—were recorded by hand on loose paper slips. Fast handwriting written during rush hours was frequently difficult for kitchen staff to read, leading to mistakes in icing colors or cake inscriptions. Loose paper slips were also easily damaged or misplaced near kitchen prep tables.
3. Manual Scheduling and Kitchen Overbooking: Delivery and pickup dates relied entirely on a paper wall calendar hanging near the kitchen prep station. Although kitchen personnel could safely decorate three to five custom cakes per day without compromising quality, counter staff taking phone orders during busy hours often failed to check the wall calendar, leading to accidental double-bookings and delayed orders.
4. Manual Bookkeeping and Calculation Errors: Downpayments, online bank transfer receipts, and daily sales totals were recorded using handheld calculators and paper logbooks. Calculating sales totals by hand after long shifts increased the risk of math errors, forcing the store owner to spend over an hour every closing shift re-checking individual paper slips.

3.1.2 Developmental Research

Developmental research focuses on using observations from the descriptive phase to design, build, test, and refine a practical software solution. Every tool and feature in the proposed web platform directly addresses a documented operational problem at C&N Bakery.

During the developmental phase, the research team translates field observations into software components across three core development areas deployed on Vercel:

TABLE 3.1 Developmental Phase Module Implementation Blueprint

Development Area	*System Module Scope*	*Planned Technical Features & Capabilities*
1. System Infrastructure & Security	*Cloud Database, Access Control & Vercel Deployment*	*• Will design cloud database schemas to store customer profiles, product catalogs, and order records*

		securely. *• Will implement Role-Based Access Control to separate administrative permissions from customer privileges.* *• Will configure automated email triggers corresponding to order confirmation, downpayment verification, and delivery/pickup status.* *• Will deploy the application on Vercel for fast, reliable cloud access.*
2. Customer Portal & Ordering System	*Customer Storefront & Interactive Checkout*	*• Will develop catalog browsing for guest users.* *• Will build an interactive Custom Cake Builder with real-time price calculations.* *• Will enforce pickup/delivery date checks and daily order limits during checkout.* *• Will build a visual Order Tracker with real-time status steps.*
3. Admin & Kitchen Dashboard	*Business Analytics & Store Control*	*• Will construct a Sales Analytics dashboard to summarize revenue trends.* *• Will implement a dual-section Order Dashboard for new and active orders.* *• Will build an interactive Schedule Calendar to toggle daily limits.* *• Will provide dynamic CRUD controls for menu items, prices, and cake choices.*

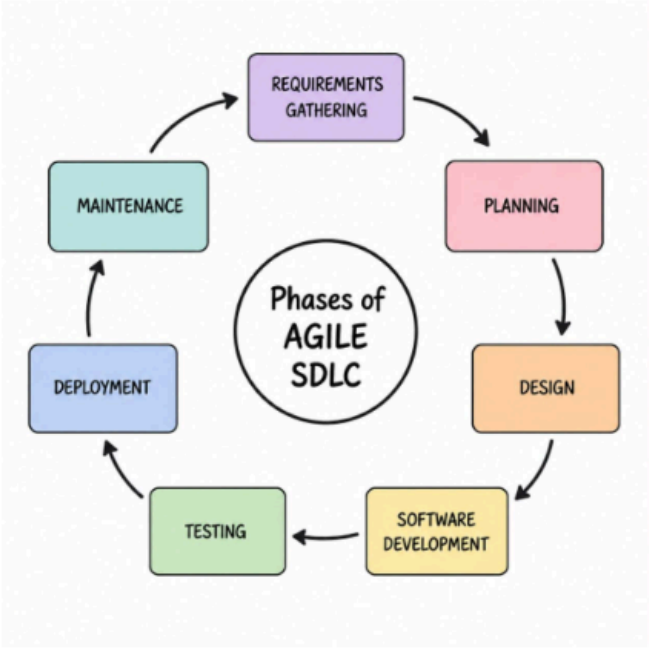
		• *Will maintain an automated activity log for system updates*

3.2 System Development Methodology

For this study, the research team selected the Agile Development Model as the primary framework for the Software Development Life Cycle (SDLC) [1]. Traditional software approaches, such as the Waterfall model, follow a rigid sequence where each phase must be finished completely before the next can begin. Under Waterfall, making changes to earlier choices after coding starts is difficult and expensive.

Developing a web-based ordering and management platform for a small business like C&N Bakery required flexibility. As the business owner, kitchen staff, and test customers interact with early site builds, practical operational details inevitably arise that were not anticipated during initial planning.

The Agile methodology will resolve this by dividing project development into short, manageable iterations called sprints [1]. Instead of waiting until the end of the project to test the entire application, the bakery owner and staff will review working feature releases early and provide direct feedback. This continuous feedback loop will reduce technical risks, prevent unnecessary feature development, and ensure the final software fits seamlessly into daily bakery operations.



3.2.1 Requirements Gathering

Concept & Purpose

The requirements gathering phase will focus on identifying the exact features, operational rules, and performance targets needed for the web platform. Setting clear feature limits early will keep development on schedule and prevent adding unnecessary tools halfway through the project.

Application to C&N Bakery Platform

During field visits to C&N Bakery, the research team interviewed the owner and shop staff to separate essential tools from features that would add unnecessary complexity. To keep the project realistic and manageable, advanced corporate tools—such as live GPS delivery tracking and automated inventory reordering—were excluded from the project scope.

From these discussions, the team grouped the system requirements into two main categories:

1. **Functional Requirements:** Core system tools defining user accounts, cake customization, cart management, capacity scheduling, administration, and reporting.
2. **Non-Functional Requirements:** Operational performance standards defining page responsiveness, data security, and loading speeds.

List of Functional Features

1. **Account Sign-Up & Browsing:** Allows customers to browse products as guests or create accounts to save addresses and view past orders.
2. **Interactive Custom Cake Builder:** Calculates cake order prices automatically as buyers select tiers, sizes, chiffon flavors, and frosting options.
3. **Order Customization Details:** Input fields for cake dedications, frosting color choices, and external design reference links.

4. **Cart & Order Progress Tracker:** A digital cart paired with a real-time progress tracker (Preparing, Baking, Packing, Ready).
5. **Smart Capacity Calendar:** Checks kitchen daily order limits and locks out fully booked pickup or delivery dates.
6. **Admin Management Portal:** A secure dashboard for bakery staff to review custom orders, approve payments, and print daily bake lists.
7. **Email Notification System:** Automatically emails customers whenever staff update their order status, payment verification, or fulfillment schedule.
8. **Sales Analytics Dashboard:** Displays simple visual charts showing daily, weekly, and monthly store revenue.

Operational Performance

Requirement Category	Performance Target	How the System Will Achieve It
Responsive Layout	Web pages will adjust cleanly across desktop computers, tablets, and mobile phones.	Front-end layouts will be built using CSS fluid grids and responsive design.
System Security	Admin management tools will be protected from unauthorized access.	Staff routes will require encrypted logins and Role-Based Access Control (RBAC).
Speed & Performance	Page load speeds will remain under fast response thresholds on both Wi-Fi and mobile data	Front-end code will be optimized, uploaded pictures will be compressed, and assets will be served via Vercel's global Content Delivery Network (CDN).

3.2.2 Planning

Concept & Purpose

The planning phase will break down system goals into manageable coding tasks and schedule them across specific development sprints. It will establish software tools, cloud backend environments, and budget constraints early to keep development organized and on schedule.

Application to C&N Bakery Platform

To keep the website fast, reliable, and free to host, the research team planned a hybrid cloud architecture combining Vercel Platform Hosting for front-end web delivery with Google Firebase Cloud Services for backend operations [3], [4]. Because local bakeries usually operate on tight budgets, keeping operational running costs at zero was set as a primary design goal. By leveraging free developer tiers across both platforms, the platform stays zero-cost through three main strategies:

1. Free Web Hosting: The website's front-end code will be deployed live using Vercel's global Content Delivery Network (CDN), providing fast rendering and built-in SSL security without hosting fees [4].
2. Zero-Cost Image & Database Storage: Firebase Realtime Database and Authentication will safely store customer orders, login accounts, and uploaded custom cake reference images under Firebase's free Spark plan [3].
3. Manual Payment Verification: To avoid paying third-party payment gateway fees, customers will pay via GCash QR codes or direct bank transfers and attach receipt screenshots. Bakery staff will verify these screenshots against the store's official communication channels before confirming the order.

Summary of Selected Software Stack

The development team selected the following key software tools based on speed, ease of use, and generous free developer limits:

- HTML5: Provides semantic web structure, accessible form fields for cake customization, and cross-browser markup standard across mobile and desktop browsers.
- CSS3:Builds responsive, fluid UI layouts and forms that adjust cleanly across smartphones, tablets, and desktop displays.
- JavaScript (ES6 OOP): Organizes website functions into reusable, secure code blocks called classes. It uses private data protection (Encapsulation), simple backend commands (Abstraction), shared card blueprints (Inheritance), and dynamic button behaviors (Polymorphism) to keep the system modular and easy to manage.
- Vercel Hosting: Deploys and serves the web application front-end interface live using Vercel's edge network for optimal loading speeds.
- Firebase Realtime Database: Syncs order updates instantly between customer tracking screens and the kitchen admin dashboard without requiring page refreshes.

3.2.3 Design

Concept & Purpose

The design phase converts feature lists into visual blueprints and structural layouts before active coding begins. Creating wireframes early allows layout adjustments on paper, saving development time during the coding phase.

Application to C&N Bakery Platform

Visual wireframes will be drawn for both the customer-facing storefront and the admin dashboard. Database structures will be outlined in the cloud database to maintain organized records:

- **Order Schema Structure:** Cake selections, design notes, image links, pickup dates, and payment statuses will link directly to a single, unique Order ID.
- **Calendar Capacity Rules:** Database tables will track active custom bookings for each date, comparing incoming requests against daily limits and locking out dates that reach maximum capacity.
- **Sales Data Schema:** Transaction totals will be stored with clear payment status tags, allowing the sales analytics dashboard to process financial metrics without slowing down the site.

3.2.4 Development

Concept & Purpose

The development phase will involve writing code, building user screens, connecting cloud backend services, and turning wireframe designs into working software. Under our Agile setup, development will be divided across three focused coding cycles called sprints [1].

Application to C&N Bakery Platform

Development will be completed across three sequential sprints, with each sprint focusing on specific platform modules:

- **Sprint 1: Customer Storefront & Custom Cake Builder**

  During Sprint 1, the research team will focus on front-end user interfaces. HTML5 markup, CSS3, and JavaScript will be written to create the user storefront. Key tasks will include building responsive product grids, coding the guest checkout workflow, and creating the interactive Custom Cake Builder with real-time price calculations.

- **Sprint 2: Admin Dashboard & Kitchen Operations Panel**

  Sprint 2 will focus on administrative and management tools for C&N Bakery staff. The team will build secure staff login routes, the dual-section Order Dashboard (New vs. Active orders), and interactive schedule controls. Code logic will be programmed to track daily order limits per date, automatically closing dates that hit maximum kitchen capacity. Simple visual charts will also be built for the sales analytics module.

- **Sprint 3: Cloud Backend Integration, Database Security, & Vercel Deployment**

  In Sprint 3, the front-end user interfaces will be connected directly to cloud database services and deployed via Vercel [3], [4]. The team will bind online form fields to cloud database paths, set up customer authentication, configure image uploads for payment receipts, and apply Role-Based Access Control security rules. Automated email triggers will also be set up to send status update alerts to customers.

3.5: Agile Sprint Development Plan

Sprint Phase	**Main Coding Focus**	**Specific Coding Tasks**	**Deliverables Produced**
Sprint 1	Customer Storefront & Cake Builder	• Will build user sign-up and login forms. • Will create product catalog grids. • Will code interactive Custom Cake Builder price calculations. • Will implement digital shopping cart management.	Working storefront, interactive cake customization form, and functional shopping cart screens.

Sprint 2	Admin Management Dashboard	• Will build secure staff login screens and role authorization. • Will program daily capacity calendar control rules. • Will build order management lists and sales analytics charts. • Will provide dynamic CRUD tools for menu pricing and cake options.	Functional admin management dashboard, interactive schedule calendar, and sales reporting interface.
Sprint 3	Cloud Integration, Vercel Deployment & Live Updates	• Will connect JavaScript user forms to cloud database paths [3]. • Will set up dynamic email notifications for order triggers. • Will configure cloud storage for receipt and cake image uploads. • Will apply role-based database security rules and deploy live on Vercel [4].	Fully connected, live web platform ready for multi-stage system testing.

3.2.5 Testing

Concept & Purpose

The testing phase will verify that all web features work correctly, remain secure, and run smoothly across different devices before public deployment. In an Agile methodology, testing will happen continuously after each development sprint to catch and fix bugs early [1].

Application to C&N Bakery Platform

The research team will execute a multi-stage testing plan across four levels:

1. **Unit Testing:** Testing individual code functions independently (such as automated price calculator math and form input checks).
2. **Integration Testing:** Making sure data flows accurately between online forms, user profile accounts, and cloud database nodes [3].
3. **End-to-End System Testing:** Running complete trial test orders from start to finish—from initial cake customization to admin order approval and sales logging.
4. **User Acceptance Testing (UAT):** Holding hands-on trial sessions with key C&N Bakery stakeholders (the owner, bakers, counter staff, and test customers) to confirm the system is easy to use.

3.2.6 Deployment

Concept & Purpose

Deployment will involve publishing the fully tested web application to live cloud hosting platforms, setting up production database paths, and preparing the software for daily business use [4].

Application to C&N Bakery Platform

The application will be published live using Vercel cloud hosting linked directly to cloud database services [3], [4]. Before opening the site to real customers, the research team will clear all test orders, sample photos, and fake

accounts from the database. The live database will then be loaded with C&N Bakery's official product menu, correct prices, product categories, and daily kitchen capacity limits.

3.2.7 Maintenance

Concept & Purpose

Following deployment, the maintenance phase will focus on keeping track of site performance, collecting user feedback during daily shop operations, and making minor adjustments to ensure system stability.

Application to C&N Bakery Platform

A post-launch maintenance plan will be implemented:

- **Performance Checks:** Database response speeds, serverless function logs, and automated calendar date locks will be checked regularly.
- **Staff Feedback Logbook:** A physical feedback logbook will be placed in the bakery kitchen for staff to write down daily operational notes or minor issues encountered during shop hours, ensuring ongoing software stability and ease of use.

3.3 System Architecture

The system architecture defines how the user interfaces, cloud database services, and external tools interact to process orders for C&N Bakery. The platform uses a client-side architecture that connects web browsers directly to cloud services, eliminating the need for expensive dedicated web servers.

3.3.1 User Browser & Web Application Layer

The web application layer handles all user interactions inside modern web browsers on mobile phones, tablets, and desktop computers. The application code is hosted on Vercel's global platform, ensuring fast page load times for both shop staff and online customers [4]. This layer is divided into two operational interfaces:

1. **Customer Storefront:** Allows customers to browse products, use the interactive Custom Cake Builder, select pickup or delivery dates, attach payment receipts, and track order progress in real time.

2. **Admin & Kitchen Management Dashboard:** Provides store owners and kitchen staff with a secure panel to review incoming orders, approve payment uploads, manage daily calendar capacity limits, update item availability, and view sales reports.

3.3.2 System Data Workflows

The system data layer manages how order details move between users and the backend database. Instead of routing requests through a traditional intermediate server, web browser client code communicates directly with Firebase Realtime Database using secure client API libraries [3].

The primary operational data flow functions as follows:

- **Order Creation:** When a customer completes a custom cake request, the web app checks daily date availability records in Firebase. If the selected date has open slots, the order details, total price, and uploaded receipt link are saved under a unique Order ID node in the database [3].
- **Real-Time Kitchen Sync:** Because Firebase Realtime Database uses active socket connections, new orders appear instantly on the kitchen dashboard screen without requiring store staff to manually refresh the browser page [3].
- **Order Status Updates:** When kitchen staff update an order status (such as moving an order from *Baking* to *Ready for Pickup*), the status change updates in the database and reflects immediately on the customer's live tracking screen [3].

3.3.3 External Integrations & Messaging Layer

To keep customers informed without requiring manual phone calls or manual text messaging during busy store hours, the architecture includes an automated messaging layer powered by EmailJS [5].

When specific order events occur in the system, automated scripts trigger client-side email requests directly to the customer's registered email address [5]:

- **Order Confirmation Alert:** Triggered immediately when a customer submits a new order request.

- **Payment Approval Alert:** Triggered when kitchen staff verify a payment receipt upload and confirm the downpayment.
- **Fulfillment Status Alert:** Triggered when an order status is updated to *Ready for Pickup* or *Out for Delivery*.

3.4 Tools and Technologies Used

This section details the software stack, development tools, cloud infrastructure, and administrative dashboards used to build, host, and maintain the C&N Bakery web platform. Each technology was selected to ensure system reliability, modern web performance, and zero operational hosting costs.

3.4.1 HyperText Markup Language (HTML5)

HTML5 provides the structural framework for all customer and administrative web pages. It utilizes semantic tags to create clear page hierarchy and accessible input forms. Standardized HTML5 form elements and validation attributes are used to collect custom cake order inputs, contact information, and delivery dates accurately.

3.4.2 Cascading Style Sheets (CSS3)

CSS3 styles the web platform and delivers a clean, responsive visual interface across all screen sizes. Using CSS fluid grid systems, flexible box layouts (Flexbox), and responsive breakpoints, the interface automatically adapts its layout when viewed on smartphones, tablets, or desktop computers. Custom CSS variables manage color themes, typography, and button states without adding extra external framework files.

3.4.3 JavaScript (JS)

JavaScript executes client-side application logic inside the user's browser using Object-Oriented Programming (OOP) principles. It manages dynamic page behaviors, such as updating prices in real time within the Custom Cake Builder, validating shopping cart items, handling image upload previews, and managing live database connections. Organizing client scripts into reusable JavaScript classes keeps system functions modular and maintainable.

3.4.4 Firebase Authentication

Firebase Authentication handles secure user management, customer sign-ups, and password logins. It encapsulates authentication processes behind secure client API calls, preventing unauthorized access to protected routes. Role-Based Access Control (RBAC) separates standard customer accounts from administrative store personnel, ensuring only verified bakery staff can access order approval panels and sales reports.

3.4.5 Firebase Realtime Database

Firebase Realtime Database functions as the cloud data engine, storing records as a structured JSON tree. Unlike traditional relational databases, Firebase uses persistent web-socket connections to sync data changes instantly. When a customer places an order or kitchen staff update an order status, the database automatically broadcasts the updated status across connected devices in real time without requiring manual browser refreshes.

3.4.6 Vercel Deployment Platform

Vercel hosts the web platform's front-end application code, delivering site assets through a global Content Delivery Network (CDN). Linked directly to the primary code repository, Vercel automatically deploys site updates whenever

new code is pushed to production. It provides built-in SSL encryption, fast domain routing, and reliable uptime without hosting fees.

3.4.7 Firebase Spark Plan (Free Tier)

To ensure the bakery incurs no monthly operational software fees, the backend operates entirely under Firebase's free Spark plan. The Spark plan includes generous daily limits for read/write database traffic, active user authentications, and cloud storage. These limits fully cover the daily transactional volume of a local bakery while maintaining zero running costs.

3.4.8 Automated Email Service (EmailJS)

EmailJS handles transactional email communications directly from client-side JavaScript without requiring a custom backend server. It connects order status triggers directly to pre-configured email templates. Automated email alerts are sent to customers upon order submission, downpayment approval, and status changes such as *Ready for Pickup* or *Out for Delivery*.

3.4.9 Visual Studio Code (VS Code)

Visual Studio Code served as the primary Integrated Development Environment (IDE) for writing, editing, and debugging source code. Armed with extensions for syntax highlighting, code formatting, and local server testing, VS Code provided an efficient workspace for building HTML5 pages, CSS3 stylesheets, and JavaScript modules.

3.4.10 GitHub

GitHub was used for source control management and source code repository hosting. It allowed the research team to track revision history, manage project branches, and merge development updates safely. Connecting the GitHub repository to Vercel enabled continuous integration, automatically deploying operational site builds upon every approved code commit.

3.4.11 Firebase Console Dashboard

The Firebase Console Dashboard serves as the cloud backend administrative portal. The research team used the dashboard to set up initial database indexing, configure security rules, inspect real-time database paths, monitor daily bandwidth usage, and manually manage user role permissions during development.

Here is the complete, human-readable draft for **Section 3.5 (Data Gathering Techniques)** written in standard academic prose, using the past tense to reflect the qualitative field research conducted during the initial problem-identification phase.

3.5 Data Gathering Techniques

To gather accurate business requirements and understand the daily operational challenges of C&N Bakery, the research team used three primary qualitative data gathering techniques: semi-structured interviews, direct field observation, and document analysis.

3.5.1 Interviews

The research team conducted semi-structured interviews with key bakery stakeholders, including the store owner, counter staff, and kitchen decorators. This approach allowed the team to ask structured questions while encouraging open conversation regarding daily work routines. The interviews helped identify major operational pain points, such as lost order notes across personal messaging apps, communication delays between counter and kitchen staff, and double-booking risks caused by manual calendar tracking.

3.5.2 Observation

The research team performed direct field observation inside C&N Bakery during regular shop operating hours and peak weekend order rushes. The team observed how walk-in and phone orders were handled, how custom cake details were transcribed onto paper order slips, and how kitchen staff cross-checked daily tasks against the paper wall calendar. These observations confirmed that manual paper workflows created physical bottlenecks, especially when multiple custom cake orders arrived at the same time.

3.5.3 Document Analysis

The research team conducted a thorough document analysis of C&N Bakery’s physical record-keeping materials. The team reviewed archived paper order slips, daily sales logbooks, printed GCash payment receipts, and the store's paper wall calendar. Analyzing these physical documents provided concrete proof of frequent handwriting readability issues, manual calculation errors in daily bookkeeping, and untracked downpayment records. These findings helped define the functional requirements for the proposed cloud database schema and admin management dashboard.

3.6 Testing Procedures

To ensure that the C&N Bakery web platform performs reliably, handles real-time orders accurately, and remains secure, the research team will execute a structured, multi-stage testing plan. Testing will occur

incrementally across four distinct phases: Unit Testing, Integration Testing, System Testing, and User Acceptance Testing (UAT).

3.6.1 Unit Testing

Unit testing will focus on verifying individual software functions and script components independently before integrating them into the broader platform. The research team will test isolated JavaScript code blocks to ensure accurate logic and error handling. Key unit tests will include:

- **Custom Cake Price Calculator:** Validating that selection math (tier sizes, chiffon flavors, and frosting add-ons) calculates correctly and updates price totals dynamically.
- **Form Input Validation:** Confirming that text fields properly enforce rules, such as checking for valid email formats, requiring phone numbers, and rejecting empty custom dedication fields.
- **Date Picker Logic:** Testing that calendar inputs correctly identify past dates and block users from picking invalid delivery times.

3.6.2 Integration Testing

Integration testing will evaluate how well individual code modules pass data back and forth when connected to cloud services. This phase ensures that front-end interface inputs sync cleanly with the Firebase cloud backend and external APIs. Key integration test workflows will include:

- **Authentication Handshake:** Verifying that sign-up forms correctly create user accounts in Firebase Authentication and assign appropriate access roles (Customer vs. Admin).
- **Database Read/Write Paths:** Confirming that submitting an order writes complete order details to the correct node in Firebase Realtime Database without dropping image links or design notes.
- **Email Triggers:** Ensuring that order status changes made in the admin portal trigger the EmailJS to deliver automated update emails to customer addresses.

3.6.3 System Testing

System testing will evaluate the entire web application as a single, fully integrated platform. The research team will perform complete end-to-end trial orders simulating real bakery operations. A typical system test run will execute the following sequence:

1. A test user creates an account, browses the catalog, and designs a custom cake using the interactive cake builder.
2. The user checks out, selects an available delivery date, attaches a sample GCash receipt screenshot, and submits the order.
3. The system locks the selected calendar date if daily capacity is reached and generates an instant automated email confirmation.
4. Bakery personnel log into the Admin Dashboard, review incoming order details and receipt attachments, approve payment status, and update production stages from *Baking* to *Ready for Pickup*.
5. The customer order tracking screen and daily sales analytics update automatically in real time without requiring manual browser refreshes.

3.6.4 User Acceptance Testing (UAT)

User Acceptance Testing (UAT) will evaluate whether the web platform meets the practical operational needs of C&N Bakery stakeholders. Hands-on testing sessions will be conducted with four primary participant groups: the bakery owner, counter staff, kitchen decorators, and sample retail customers.

Participants will complete common daily tasks using the platform without assistance from the research team. Feedback regarding ease of use, interface readability, and task efficiency will be collected using a standardized evaluation survey to verify that the platform is ready for daily business deployment.

3.7 Evaluation Criteria

To evaluate the overall software quality and practical performance of the web platform, the research team will adopt the ISO/IEC 25010 Software Quality Model [2]. The system will be evaluated by two target groups: IT software experts and end-user stakeholders (including the bakery owner, shop personnel, and sample customers).

Evaluators will rate the application using a standardized 5-point Likert Scale instrument ranging from **5 (Strongly Agree)** to **1 (Strongly Disagree)** across five core quality characteristics [2]:

3.7.1 Functionality

Functionality evaluates whether the web platform provides all required operational features and accurately executes tasks according to system specifications [2]. Evaluators will assess whether the Custom Cake Builder calculates prices accurately, the Smart Capacity Calendar blocks overbooked dates, the Admin Dashboard processes status updates, and automated email notifications deliver reliably.

3.7.2 Usability

Usability measures how easily users can learn, navigate, and interact with the platform interface [2]. Evaluators will assess visual layout clarity, button responsiveness, form accessibility, and overall operational simplicity across both mobile smartphones and desktop screens.

3.7.3 Reliability

Reliability assesses the stability, consistency, and error tolerance of the platform during daily store operations [2]. Evaluators will test whether the web application maintains stable cloud database connections, prevents data loss during checkout, and operates without unexpected system crashes.

3.7.4 Efficiency

Efficiency measures system performance, page rendering speeds, and resource utilization [2]. Evaluators will measure how fast web pages load under standard mobile network speeds, how quickly real-time order

updates sync between customer screens and the admin panel, and how efficiently uploaded receipt images are processed.

3.7.5 Security

Security evaluates data protection, account privacy, and access control mechanisms [2]. Evaluators will verify that user passwords are encrypted, administrative routes are protected behind Role-Based Access Control (RBAC), customer personal details remain private, and payment receipt files are accessed safely.

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CHAPTER IV

RESULTS AND DISCUSSION

4.1 Experimental Setup and Testing Environment

The Custom Cake Ordering Platform was evaluated across multiple client and server deployment environments to assess its system performance, responsiveness, database synchronization speed, and operational efficiency. Testing was conducted using the primary administrative workstation alongside representative customer mobile devices.

The test environment configuration included:

Admin Workstation	12th Gen Intel® Core™ i5-12450H (2.00 GHz), 8 GB RAM, 512 GB SSD, Full HD (1920 × 1080) display running Windows 11 Home (64-bit).
Network Environment	Fiber optic connection with a sustained throughput of 50 Mbps.
Backend Database	Google Firebase Realtime Database https://console.firebase.google.com/project/candn-bakery/database/candn-bakery-default-rtdb/data
Hosting Platform	Vercel Global Edge CDN for automated deployment and HTTPS distribution.
Client Devices	Mobile devices running Android 13 and iOS 16, accessed via Google Chrome and browsers.

4.2 Functional Verification and Performance Analysis

The platform's core functional modules—including the Custom Cake Builder, Smart Scheduling Engine, 4-Stage Kitchen Tracker, and Sales Analytics Dashboard—were evaluated using simulated user actions and operational store workflows.

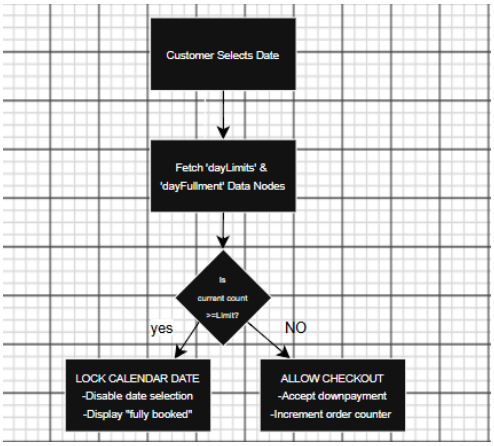
1. Configurator Input ● Tiers, sizes, flavors ● Custom text & uploads ● Instant price arithmetic	2.Smart Capacity Validation ● Checks 'dayLimits' node ● Validates 'dayFulfillment' state ● Auto-locks locked/full dates
4.Sales Analytics Engine ● Updates revenue graphs ● Logs category metrics ● Tracks popular products	3. 4-Stage Kitchen Tracker ● Pending -> Confirmed -> Baking -> Ready ● Instant Firebase sync ● Automated status emails

4.2.1 Custom Price Calculation Accuracy

The automated price calculation module was evaluated by inputting various multi-tier custom cake configurations and comparing the output against C&N Bakery's manual pricing formulas. Across all tested combinations (spanning varying cake tier counts, chiffon flavor bases, specialty frostings, and decorative toppers), the system calculated price totals instantly in $< 1second$ with 100% mathematical accuracy. Furthermore, the system accurately enforced the mandatory 50% deposit rule at checkout.

4.2.2 Smart Scheduling & Dynamic Slot Control

The dynamic capacity controller was evaluated by submitting consecutive orders targeted for specific pickup/delivery dates. When order submissions reached the threshold set in the dayLimits collection node, the system updated the dayFulfillment node in real time. As a result, the date picker interface in the customer checkout window instantly disabled and locked the fully booked calendar dates, preventing kitchen overbooking.



4.2.3 Kitchen Tracking and Real-Time Synchronization

The 4-stage kitchen tracker was evaluated by transitioning orders through production stages (*Preparing* → *Baking* → *Packing* → *Ready for Pickup/Delivery*) on the Admin Panel. Order status updates synchronized to the customer's live tracking view within $< 2 seconds$ via Firebase Realtime Database websockets. Automated email notifications were successfully dispatched to customers upon each stage progression.

4.3 Operational Efficiency Metrics and Benchmarks

To quantify operational improvements, timing benchmarks were recorded comparing traditional manual paper-based methods against the developed web platform.

Processing Time Comparison (Manual vs. Developed System)

Custom Price Calculation	Manual: 5-10 mins System: <1 sec
Capacity Check	Manual: 2-5 mins System: <1 sec
Monthly Sales Reporting	Manual: 1-2 hours System: Instant

4.4 Discussion

The empirical results demonstrate that transitioning C&N Bakery from manual logbooks to a centralized cloud architecture resolves several operational vulnerabilities:

1. **Elimination of Custom Order Ambiguity:** By using structured input forms in the Custom Cake Builder, essential design parameters (such as layer specifications, frosting choices, text dedications, and reference images) are bound directly to a single orderId node. This eliminates order transcription errors caused by informal social media messaging threads.
2. **Prevention of Kitchen Overbooking:** The automated capacity-checking logic dynamically locks dates in the customer-facing calendar. This protects kitchen staff from over-committing daily labor capacity during peak operational periods.
3. **Enhanced Data Security and Business Intelligence:** Moving data to Google Firebase Realtime Database eliminates physical loss risks associated with paper receipts. Simultaneously, the automated analytics module compiles sales histories into clear visual charts, providing management with direct insights into top-selling products and revenue trends.

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CHAPTER V

CONCLUSION AND RECOMMENDATIONS

5.1 Introduction

This chapter presents the summary of findings, overall conclusion, and practical recommendations for the C&N Bakery platform. Throughout this project, the team focused on solving the main operational struggles of C&N Bakery—specifically managing custom cake details, preventing overbooking in the kitchen, tracking order progress, and viewing sales reports.

This chapter brings together the results from Chapter 4, reviews how well the project goals were met, and shares helpful recommendations for future developers and business owners. By combining online cake ordering, live tracking, automatic price calculations, calendar limits, and sales charts into one easy platform, this study shows how simple web tools can streamline daily operations for a local bakery.

5.2 Summary of Findings

Testing and implementing the C&N Bakery platform provided clear results that directly match the goals set at the beginning of the project:

- Automatic Pricing and Faster Ordering: The website successfully replaced messy paper order slips and disorganized social media chats. Prices for custom cakes are calculated automatically the moment a customer picks their cake size, tiers, frosting, and design add-ons. This cleared up price confusion, removed math errors, and made ordering way faster for customers.
- Fewer Kitchen Mistakes on Cake Designs: The custom cake builder stopped design mix-ups by letting customers submit exact details—like cake sizes, chiffon flavors, frostings, custom messages, and reference photo uploads—right on the site. Baking staff no longer have to guess or misread handwritten notes, which cut down on wasted ingredients and rejected cakes.
- Easy-to-Find Store Rules and FAQs: Adding an FAQ page with clear store policies, advance ordering rules, and payment steps gave customers quick answers to their questions. This saved front-desk staff from answering the same messages over and over again.
- No More Overbooked Kitchen Days: By setting daily order caps in the database, the site checks calendar availability before taking an order. During testing, trying to pick a fully booked date blocked the order and asked the customer to pick a different day. This completely fixed the problem of taking on too many orders during busy seasons.
- Clear Order Status for Kitchen and Customers: The 4-stage kitchen tracker (*Preparing* → *Baking* → *Packing* → *Ready*) gave bakers a clear task list and gave customers direct visibility into their order progress. Customers didn't need to call or text the store for updates anymore because updating a stage on the admin panel instantly updated the customer's screen.
- Organized Pickup and Delivery Times: Forcing customers to pick a specific date and time slot for pickup or delivery stopped scheduling mix-ups. Bakers now know exactly when a cake is needed, making sure orders are baked fresh right before pickup instead of sitting around or being forgotten.
- Safer Customer Records and Cloud Backups: Moving away from paper notebooks protects customer phone

numbers, home addresses, and order histories from being lost, spoiled by kitchen spills, or seen by unauthorized people. Everything is stored safely in cloud database backups with secure logins for staff.

- Instant Sales Numbers and Customer Reviews: The visual sales dashboard turns completed orders into simple revenue charts, popular product lists, and date-filtered reports. This saved the bakery owner from spending hours manually adding up paper receipts at the end of the month. Plus, the built-in rating feature gave the owner direct feedback on cake quality and service straight from customers.
- Easy to Learn and Use: Testing the site with bakers, store staff, and real customers proved the layout is clean and simple. Staff picked it up quickly without needing long training sessions, and user mistakes dropped significantly.

5.3 Conclusion

The custom ordering, tracking, and management platform built for C&N Bakery successfully achieved every single goal set for this project. It delivers a practical, easy-to-use digital tool created specifically around the day-to-day needs of a growing local bakery.

Before this system was built, C&N Bakery struggled with lost paper order slips, mixed-up cake design details, accidental double-booking during peak seasons, and time-consuming manual math for monthly sales. The new web application completely solved these operational headaches. By moving store management to a central cloud database, custom cake prices are calculated automatically, daily kitchen limits are enforced strictly, and baking schedules run much more smoothly.

On top of that, live order tracking built real trust with customers and drastically cut down on constant status check messages. The admin panel simplified store operations by giving staff straightforward control over menu items, calendar booking caps, store settings, and instant sales reports.

In short, this project proves that accessible, well-designed web tools can transform small food businesses. The C&N Bakery platform successfully bridges the gap between everyday kitchen work and smart digital management—improving productivity, preventing costly mistakes, and giving the bakery owner clear numbers to make confident business decisions.

5.4 Recommendations

To build on what we've accomplished and keep the website useful for C&N Bakery as the business grows, we recommend the following system updates, research ideas, and daily store habits:

PROPOSED SYSTEM

SYSTEM IMPROVEMENTS	FUTURE RESEARCH	OPERATIONAL STRATEGY
- SMS Text Alerts **- Direct Payment**	**- Native Mobile App** **- Long-Term Revenue**	**- Regular Staff Train** **- Database Backups** **- Customer Reviews**

5.4.1 System Improvements

- Direct Payment Gateway Integration: Future developers should consider integrating automated payment gateways (such as PayMaya) directly into the checkout pipeline. This would enable instant payment verification without requiring staff to manually check payment screenshots sent via Facebook Messenger.
- SMS Text Alerts: While the current site sends email updates, adding direct SMS text messages would alert

customers the second their cake is ready for pickup or out for delivery.

5.4.2 Future Research Directions

- Long-Term Revenue Study: Running a 6-to-12-month study would help measure the system's long-term effect on store profits, staff working hours, and repeat customer rates.
- Dedicated Mobile App: Building a standalone mobile app for Android and iOS could give customers offline order drafts and better push notifications on their phones.

5.4.3 Implementation Strategies

- Comprehensive Staff Training: C&N Bakery management should run regular training sessions for all kitchen and front-desk staff. Consistent data entry habits such as updating order stages immediately as work happens in the kitchen are essential to keep customer tracking accurate.
- Routine Database Backups and Security Audits: System administrators should schedule regular database backups and periodically update security rules within Firebase to safeguard customer data and store records.
- Regular Customer Feedback Reviews: Business managers are encouraged to check the integrated reviews node regularly to spot quality trends, address customer concerns quickly, and keep product offerings aligned with customer preferences.

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